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Agentic AI in Software Engineering

From Data to Solutions: Weekend 6 – Agentic AI

Piyushi Goyal

Where this lecture sits in the arc

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Part 1: From autocomplete to agents

The three generations

2021
Autocomplete

Suggests the next line while you type.

GitHub Copilot

You watch, you accept.

2023
Chat

You ask, it answers, you paste.

ChatGPT · Copilot Chat

You ask, you paste.

2024–2026
Agents

You are "***vibe-coding***" – describe the goal and let the AI handle rest!

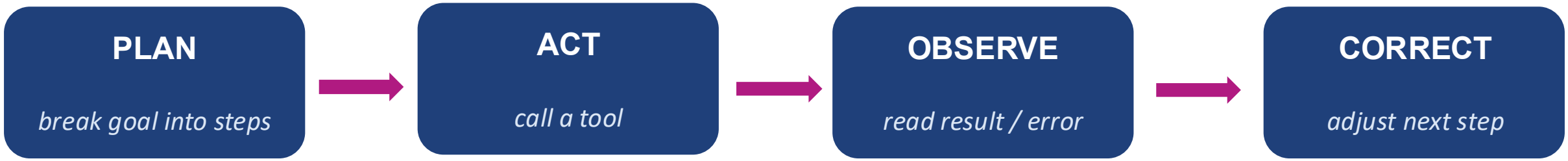
Cursor · Claude Code · Devin

You describe a goal, you review.

ScoutAI lives in the third column.

An agent = LLM + tools + a loop

An agent is a language model given tools and a loop that lets it act, observe, and correct.



Loop until the goal is met (or a stop condition)

Analogy

- *A chatbot is **Siri***
- *An agent is **JARVIS**: Tony says "run diagnostics on the suit and reorder the palladium", and by the time he lands, it's done.*

1. https://rdi.berkeley.edu/llm-agents/assets/llm_agent_history.pdf

When to reach for it: workflow vs. agent

WORKFLOW = fixed pipeline

- Use when steps are known in advance.
- Cheaper, more predictable, easier to test.

AGENT = steps chosen at runtime

- Use when the model must choose its next step based on what it just saw.
- More powerful, harder to audit, costs more per run.

Rule of thumb: if you can draw the pipeline on a whiteboard, build a workflow. Only reach for an agent when you can't.

1. https://www.anthropic.com/engineering/building-effective-agents

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Part 2: Building ScoutAI

Meet ScoutAI — the app we'll build together

"Give me a one-page brief on Competitor X."

The old way

2 days of a junior analyst's time.

- Read the annual financial reports
- Scrape the website for pricing
- Check LinkedIn for hiring signals
- Pull the latest news headlines
- Write the memo

With ScoutAI

~4 minutes.

- Type: analyse("Competitor X")
- Agent plans, fetches, drafts
- Live prices from yfinance
- Real filings from SEC EDGAR
- Human reviews and signs off

Under the hood — where the LLM comes from

OpenRouter
one endpoint, many models

HuggingFace
open-weights, self-host

Anthropic, OpenAI, Gemini
API-based frontier models

AWS Bedrock / Azure Foundry / Google Vertex AI
cloud-native gateways

Start here: a plain LLM hits a wall

"One-page brief on NVIDIA. Live price, last week's news, risk factors."

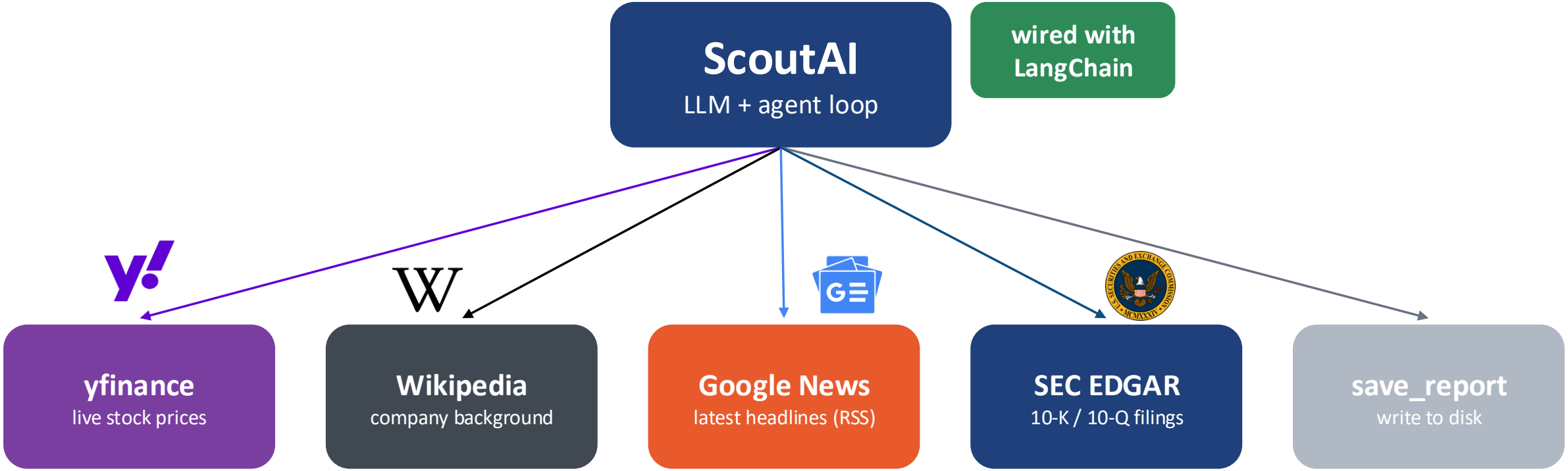
A chatbot alone cannot do this.
Here's why →

Five gaps a plain LLM leaves

- **Stale price**
  training data ends months ago
- **Hallucinated news**
  confident, plausible, wrong
- **No sources**
  you can't verify a single claim
- **Drifting format**
  each run looks different
- **No memory**
  yesterday's notes vanish

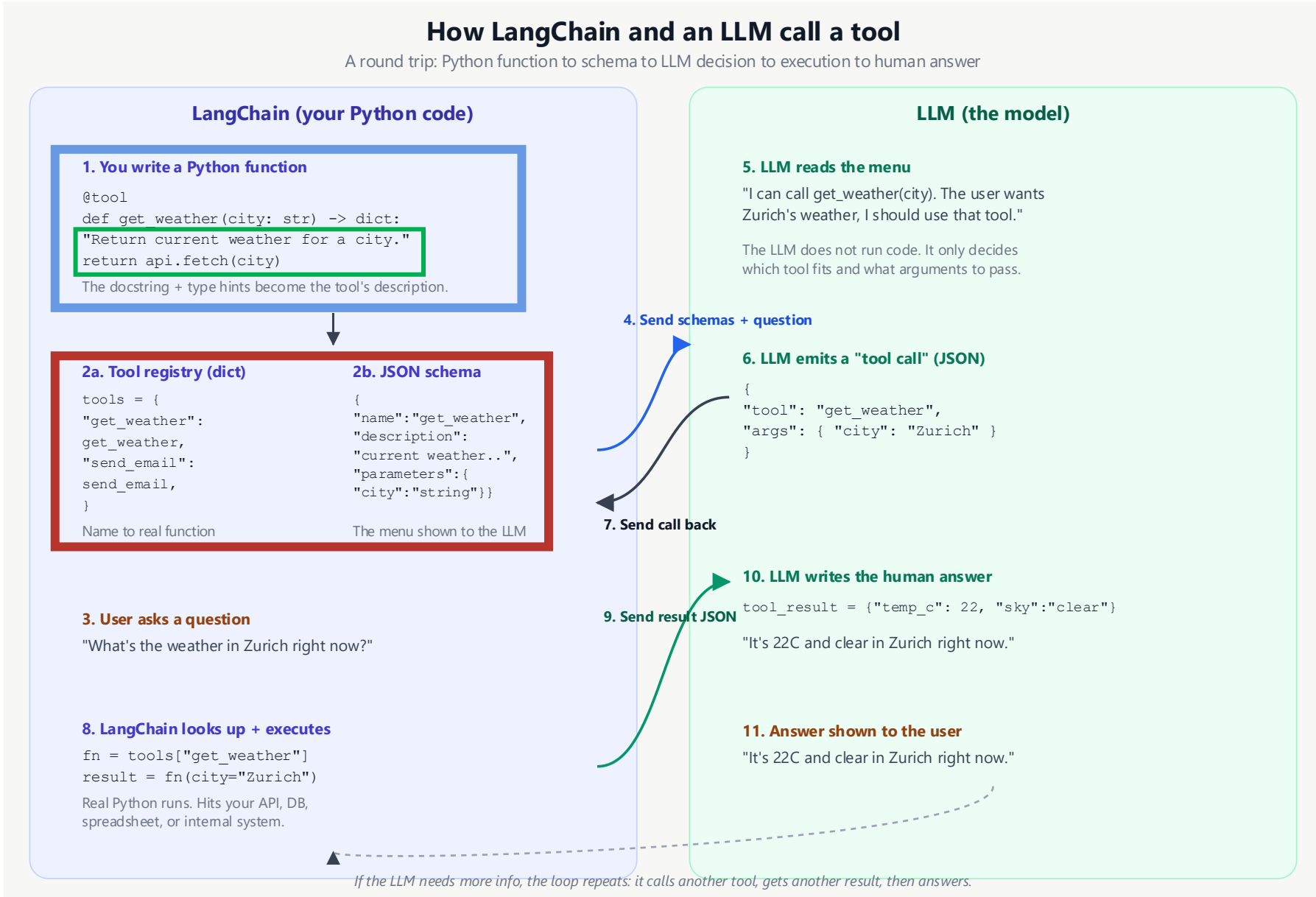
Tools: the agent's hands

A tool = any function the LLM is allowed to call.
It can be an API call, a shell command, a database query, a calculator; anything that lets the model act beyond text.

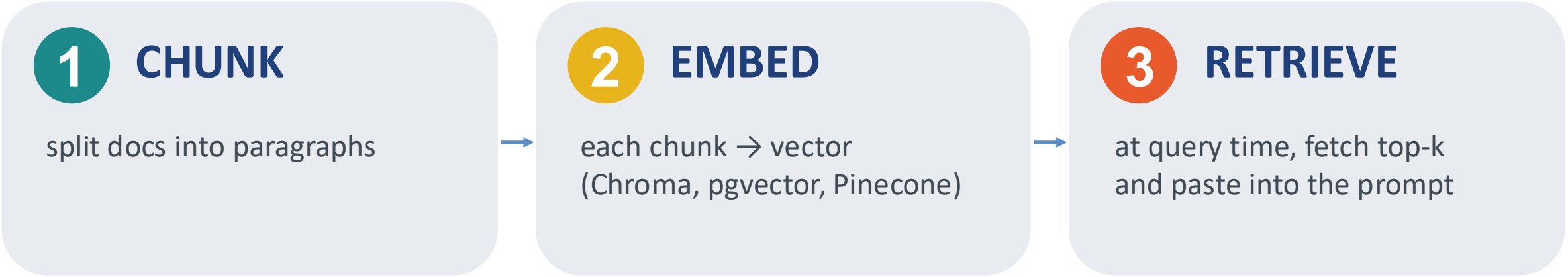


In the CX, every one of these five hits a real, live API.

Tools: LangChain and LLM



RAG: giving the agent your private knowledge



In ScoutAI:

Five internal memos (channel checks + house rules) are chunked, embedded in Chroma, and exposed to the agent as a search_our_research tool.

The agent decides on its own when to call it — no if-statement.

RAG = unstructured knowledge (docs). Tools = live data and actions. Use both.

Skills: reusable playbooks the agent invokes

What Skills look like

A Skill directory:

```
anthropic_brand/
    ├ SKILL.md
    ├ docs.md
    ├ slide-decks.md
    └ apply_template.py
```

anthropic/brand_style/SKILL.md

YAML Fontmatter

```
---
name: Anthropic Brand Style Guidelines
description: Applies Anthropic's official brand colors and
typography to PowerPoint presentations
---
```

Metadata

Markdown

```
## Overview

This skill provides Anthropic's official brand identity resources for
PowerPoint presentations. It includes a pre-branded template
and tools to apply Anthropic styling to existing presentations.

## Colors

*Main Colors:*

- Dark: `#141413` - Primary text and dark backgrounds
- Light: `#faf9f5` - Light backgrounds and text on dark
- Light Gray: `#e8e6dc` - Subtle backgrounds

### Typography

- *Headings*: Poppins (with Arial fallback)
- *Body Text*: Lora (with Georgia fallback)
```

Freeform Content

Without same prompt → different-looking report each run.

With identical structure. Auditable, reviewable, versionable.

2025-26 Anthropic, Cursor, OpenAI all ship Skills as first-class.

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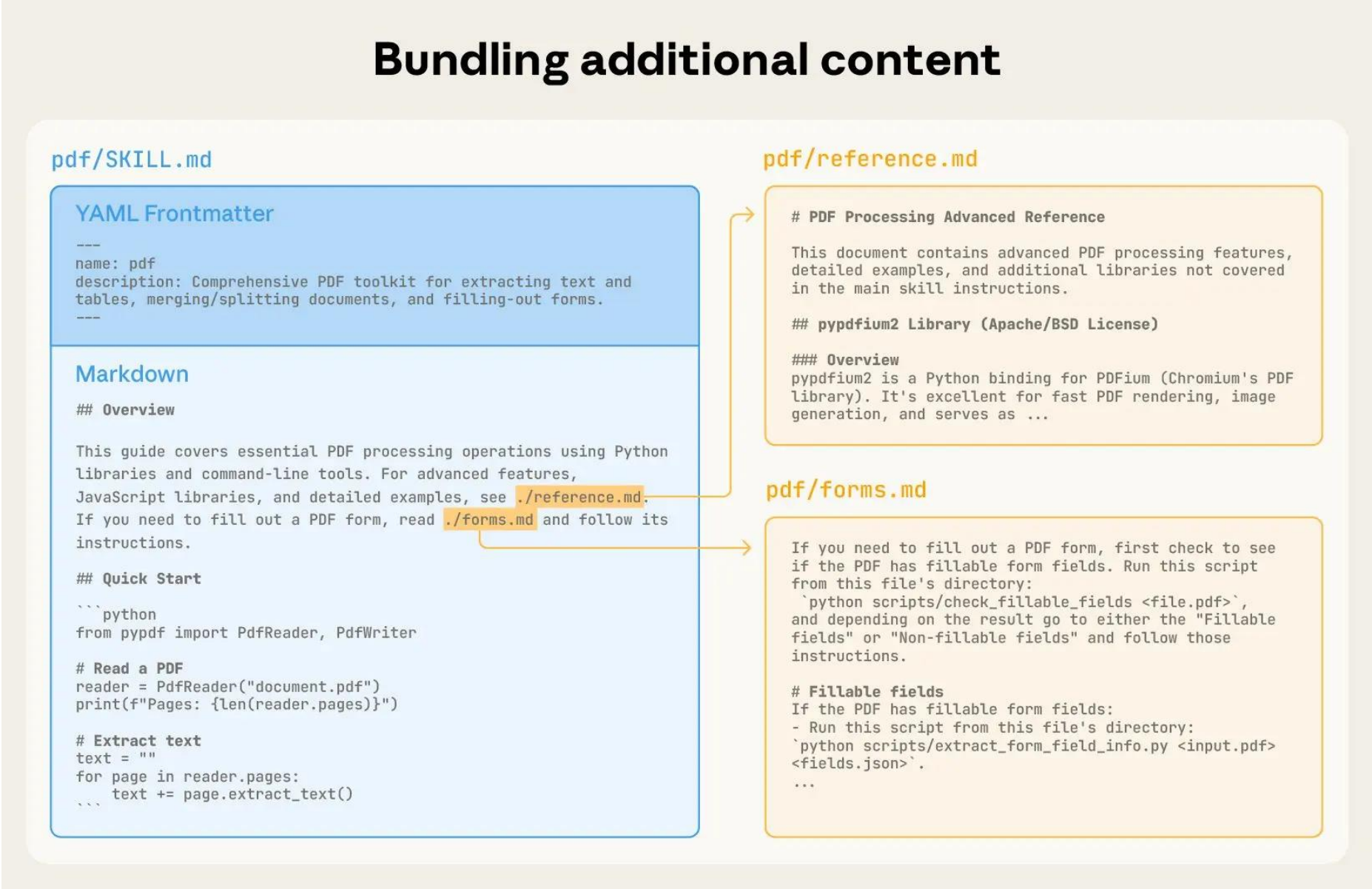
https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills

Skills: reusable playbooks the agent invokes



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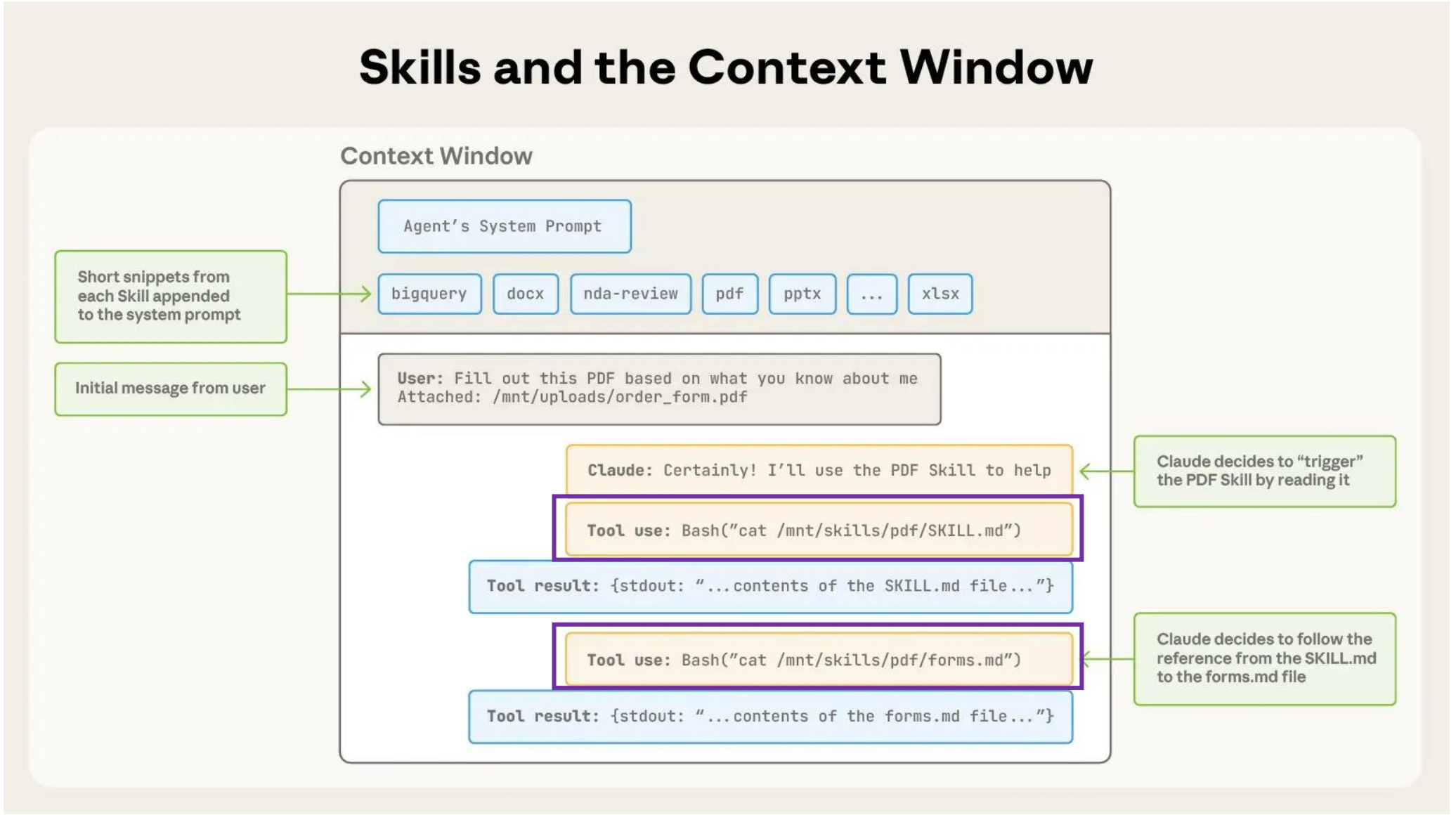
Skills: reusable playbooks the agent invokes



https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills

Memory: what the agent remembers

SHORT-TERM · context window

USER — Analyse Competitor X

SCOUT — Reading 10-K...

SCOUT — Q3 revenue: $2.1B

SCOUT — Retrieved 4 quotes on margin

SCOUT — Drafting section 2 of 5...

✕ Gone when the session ends.

LONG-TERM · persistent memory

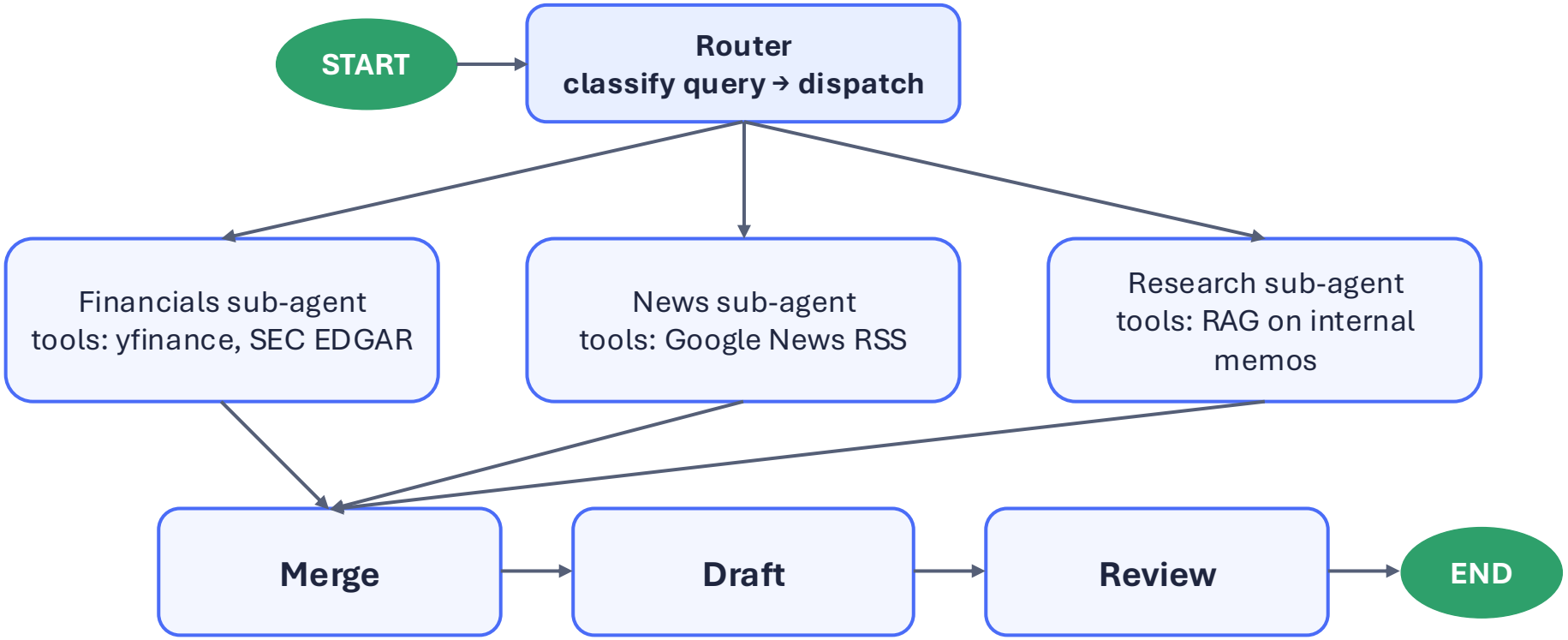
Vector DB / KV store · Mem0, LangMem, Zep, Memory.md

- House style: prefer bullets.
- We covered Comp X in June — reuse.
- CFO wants risk section first.
- Never quote unpublished figures.
- Tone: cautious, hedged.

✓ Survives across sessions and across agents.

LangGraph: an orchestrator that dispatches sub-agents

The orchestrator doesn't do the work. It routes to specialist sub-agents, each with its own tools and system prompt.

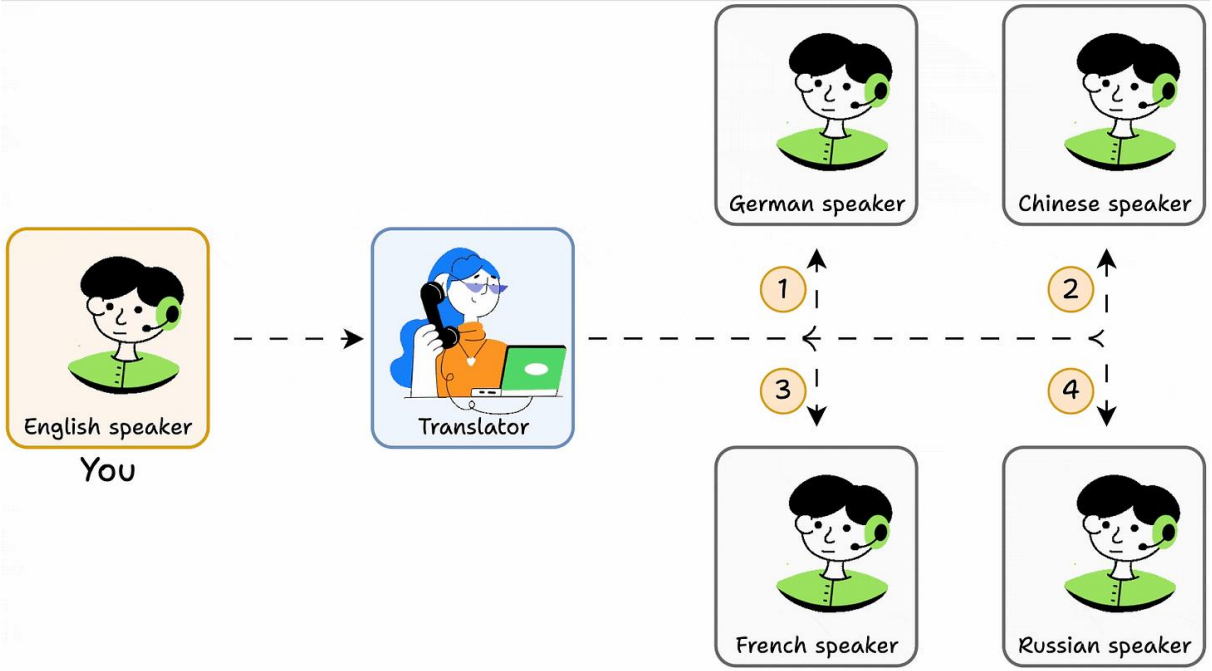


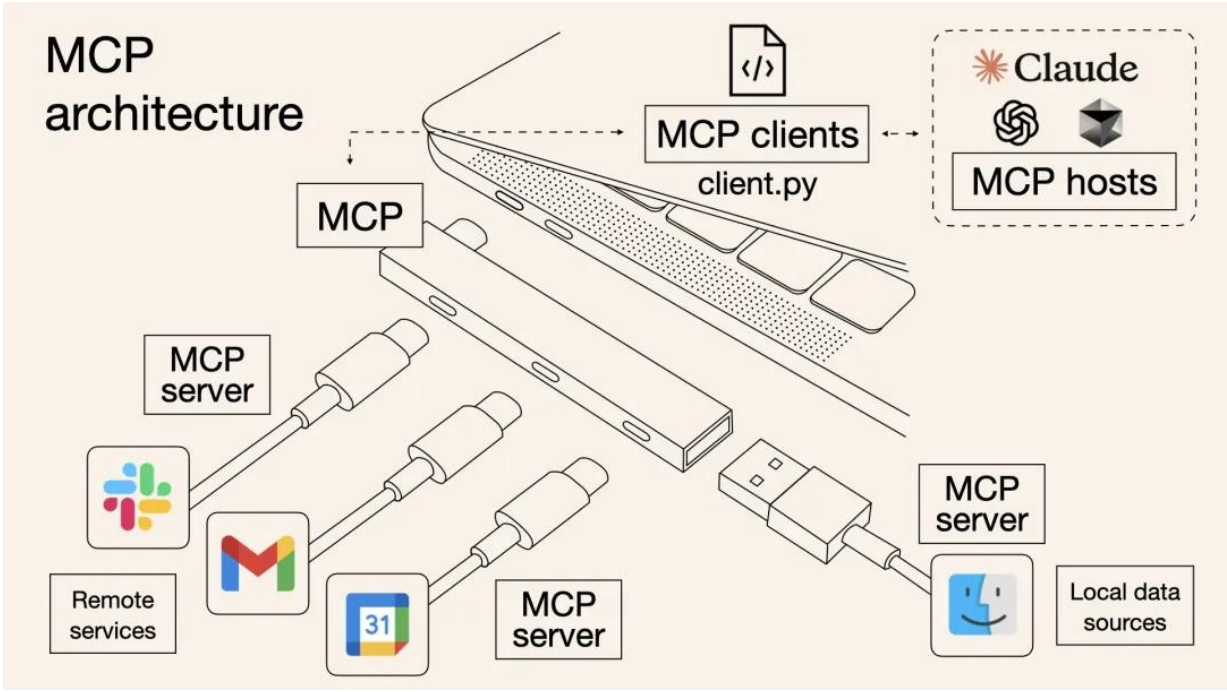
Uber

CISCO

Reach for it whenever the workflow needs branches, loops, or multiple agents; draw the graph first, then wire the nodes.

Model Context Protocol (MCP): acts like a USB-C





Open-sourced by Anthropic, Nov 2024. Backed by OpenAI, Google, Microsoft.

The single most important protocol name to know. Now backed by Anthropic, OpenAI, Google, Microsoft.

https://www.dailydoseofds.com/p/visual-guide-to-model-context-protocol-mcp/
https://www.dailydoseofds.com/p/hands-on-building-an-mcp-server/

MCP architecture: host · client · server

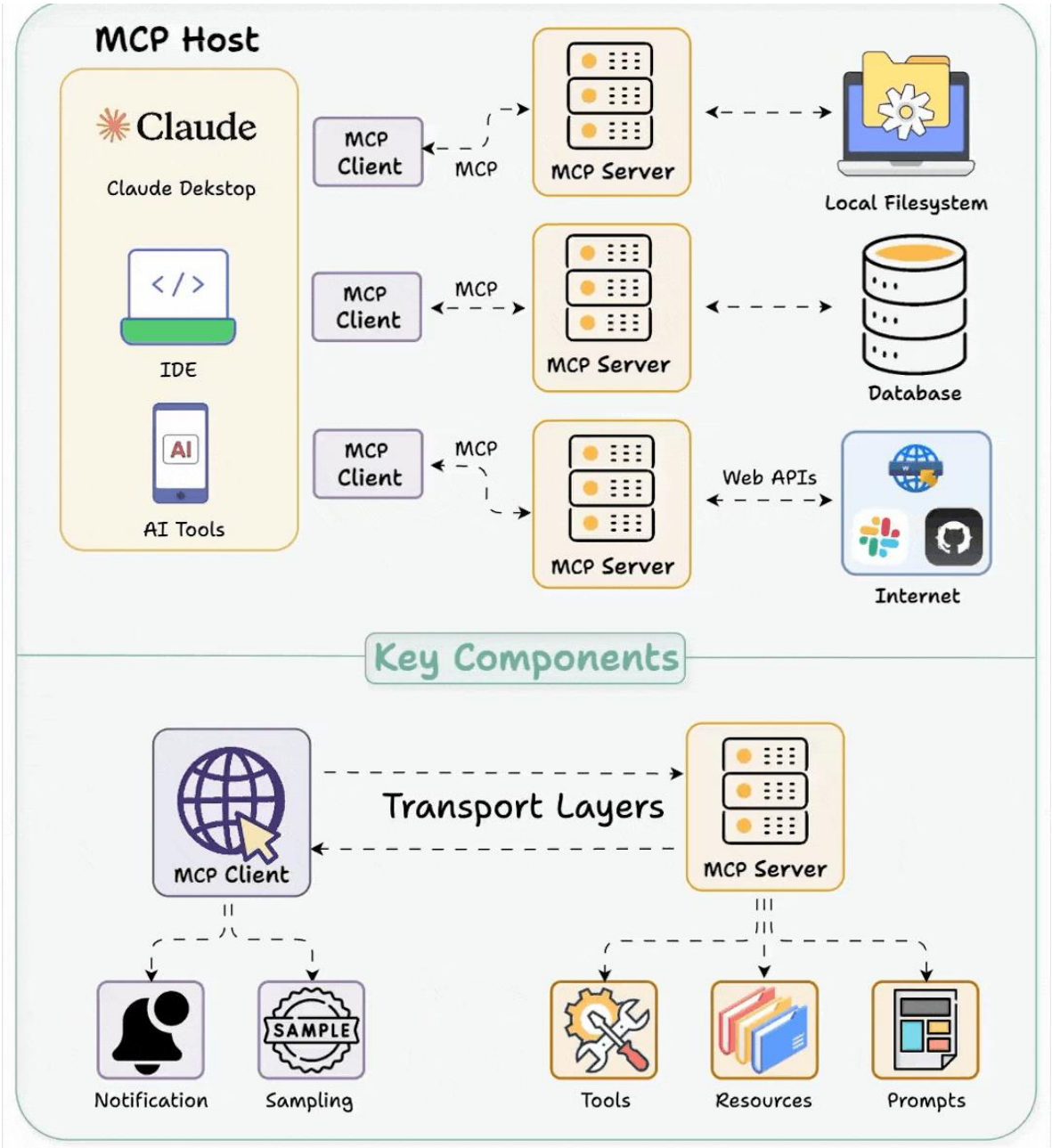
HOST — user-facing app (Claude Desktop, Cursor, ChatGPT, your agent)

CLIENT — speaks MCP; lives inside the host (one per open server)

SERVER — wraps an app or data source (GitHub, Postgres, Slack, your API)

stdio · local process, no network hop (filesystem, terminal, local apps)

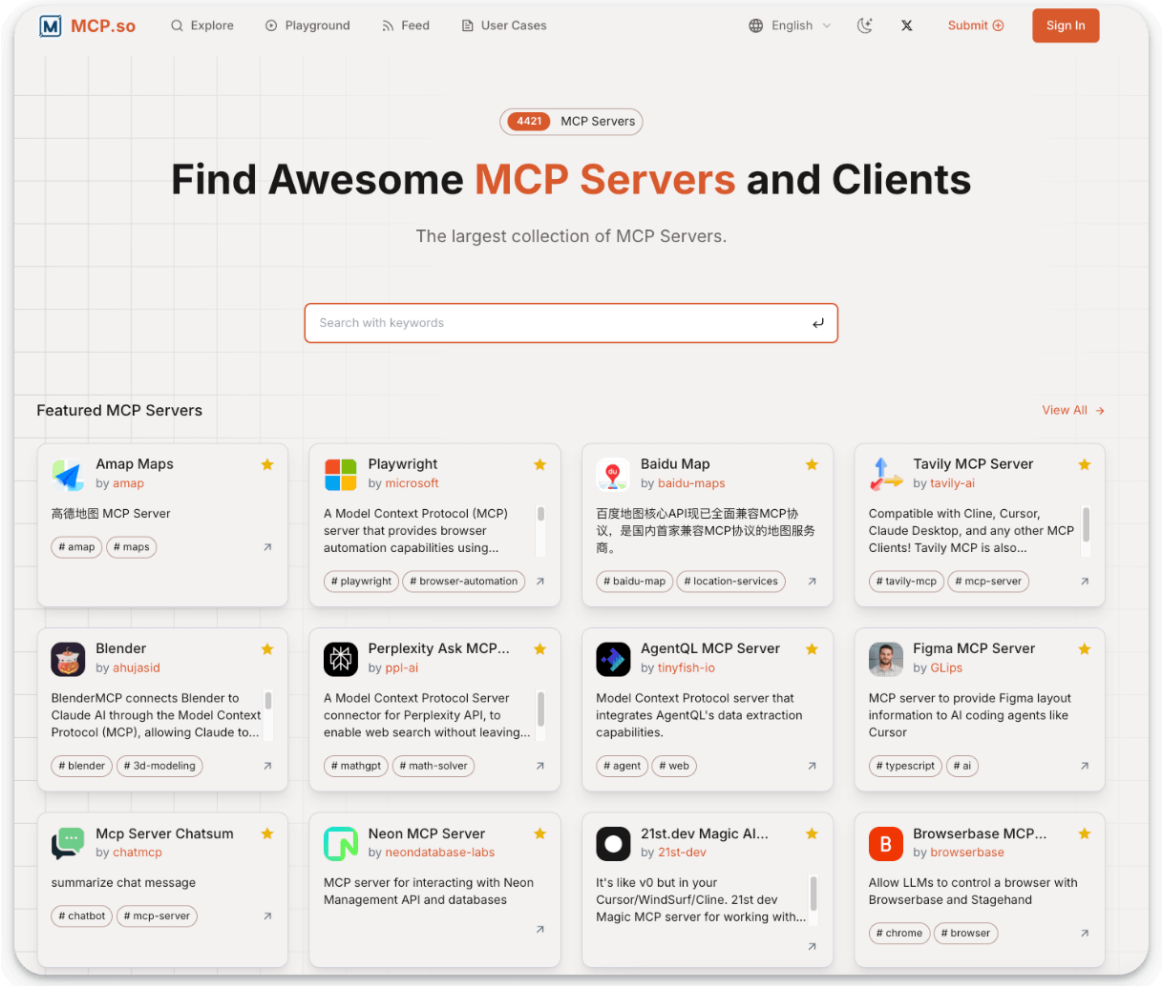
HTTP / SSE · remote server, org-wide (Slack, Github, Notion, Saleforce)



https://www.dailydoseofds.com/p/visual-guide-to-model-context-protocol-mcp/

Where servers and clients come from

1. MCP Registry
whole apps as tool bundles



2. PyPI / npm
traditional package registries

```
pip install langchain-slack
```

```
npm install @langchain/openai
```

500+ LangChain connectors

3. Your own
company-internal tools

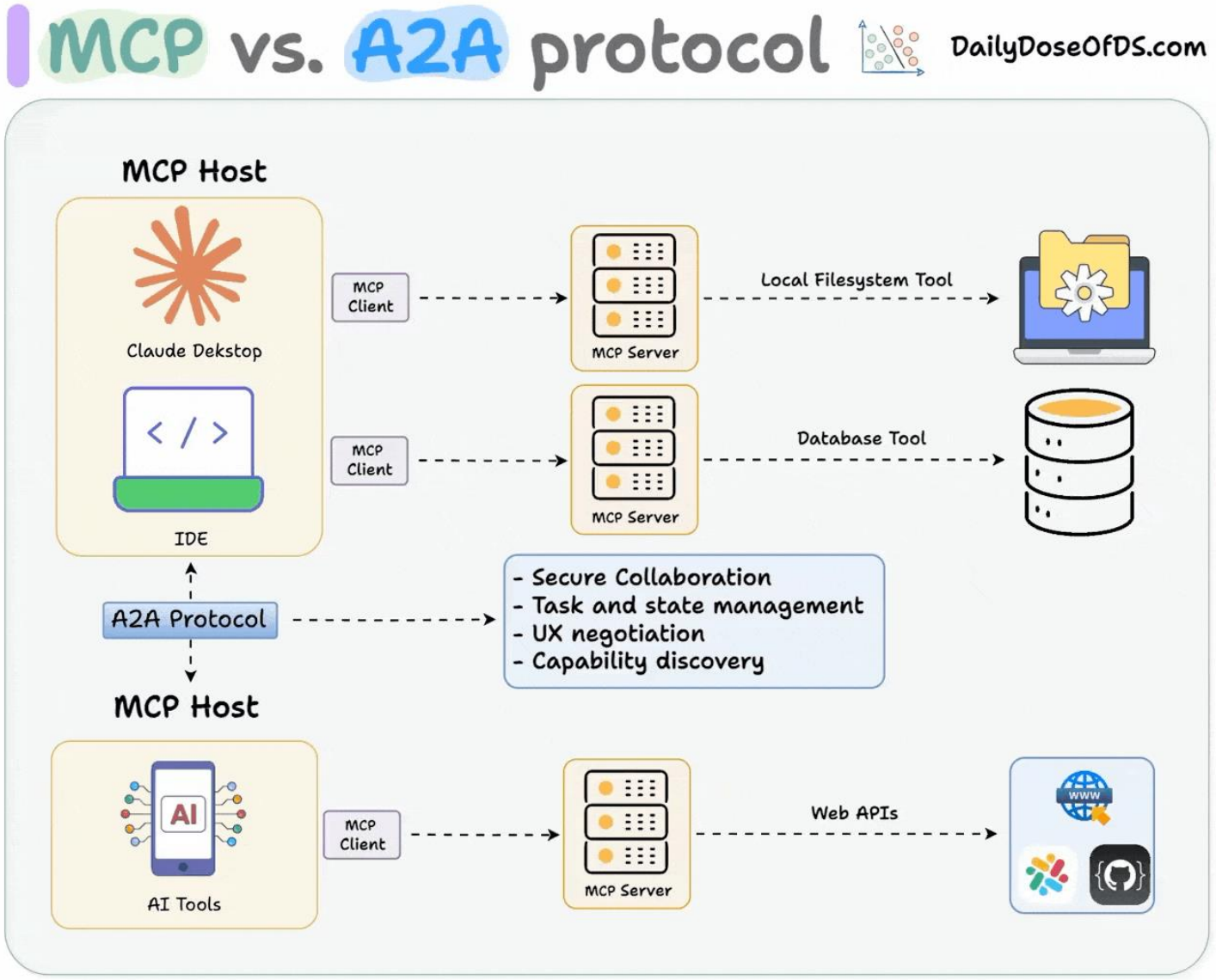
```
query_our_datawarehouse
```

```
post_to_our_pagerduty
```

```
check_our_compliance_rules
```


https://mcp.so

Agent to Agent (A2A) protocol: agents talking to agents



Think Nick Fury

Fury doesn't fight aliens himself. He calls the Avengers.

Orchestrator agent = Fury.
Specialist agents = Avengers.

MCP = each hero's gear. A2A = the comms earpiece.

https://blog.dailydoseofds.com/p/a-visual-guide-to-agent2agent-a2a

A2A: agents talking to agents

How agents are discovered

Direct URL
Private registry
Public catalog

```
// A2A envelope
{
  "id": "task-4711",
  "role": "agent",
  "task": "review brief for MNPI",
  "input": { "brief": "NVDA Q3 memo" },
  "status": "streaming"
}
```

Guardrails

Authentication
Capability scoping
Task-level permissions
Human-in-the-loop

Agent Card
public JSON: what I can do, my endpoint, my auth

Task
unit of work with an ID and streamed status

Message
role-tagged

https://developers.googleblog.com/en/a2a-a-new-era-of-agent-interoperability/

A2A in the wild

Google Cloud: hiring workflow

Hiring manager's agent → sourcing agent → background-check agent. Three agents, three vendors, one job.

Link to the demo!

Salesforce Agentforce ↔ Google Agentspace

A Salesforce service agent delegates to a Google-side agent for Workspace context. Zero custom integration.

SAP Joule & ServiceNow Now Assist

Both platforms adopted A2A so their agents interoperate outside their own walls.

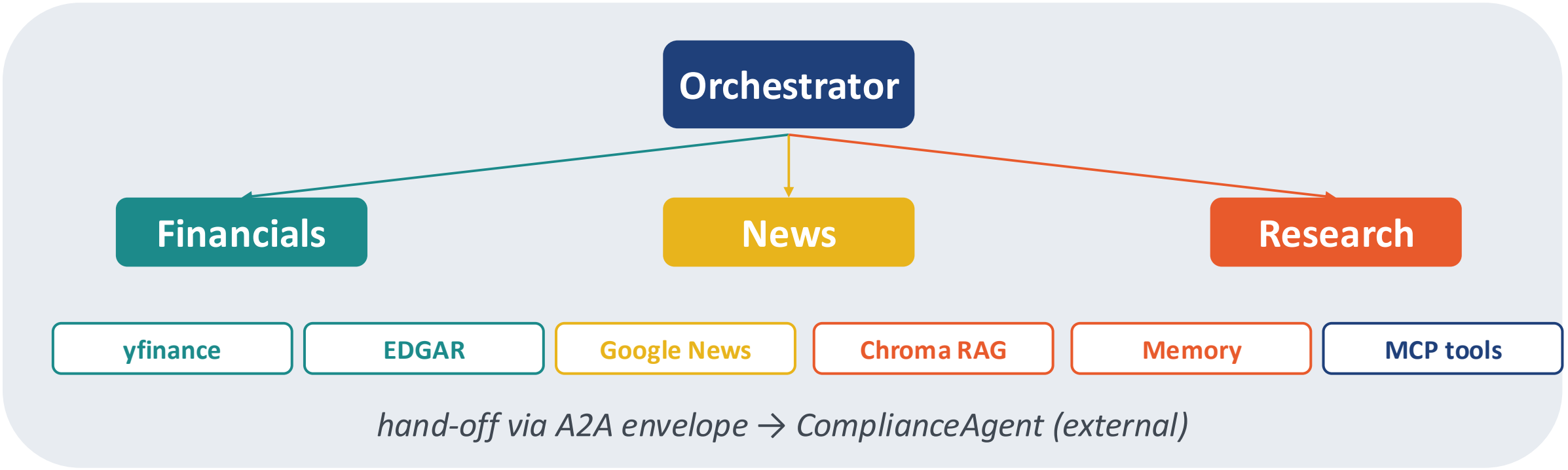
Enterprise pilots: Deloitte, Accenture, BCG

Coordinating agents across their internal stack and their clients' stacks.

https://cloud.google.com/blog/products/ai-machine-learning/google-agentspace-enables-the-agent-driven-enterprise
SAP Article
https://ikala.ai/blog/ai-trends/an-in-depth-analysis-of-googles-a2a-protocol-and-its-relationship-with-anthropics-mcp-en/

ScoutAI, fully assembled

LLM 1 | Tools 2 | RAG 3 | Skills 4 | Memory 5 | LangGraph 6 | MCP+A2A 7



The CX builds this from left to right, one concept per part.

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Part 3: Agents across the SDLC

The whole lifecycle

Same seven primitives from Part 2. Different stage, same machine.

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6
Reqs & Design	**Coding**	**Code Review**	**Testing & QA**	**Debug & Incidents**	**DevOps & Deploy**
Draft user stories, PRDs, acceptance criteria	Multi-file edits, refactors, feature implementation	First-pass PR review, diff summaries, security flags	Unit tests, E2E flows, edge-case fuzzing	Log triage, hypotheses, hotfixes, post-mortems	Terraform / Helm / CI, dependency bumps
TOOLS YOU'LL HEAR	**TOOLS YOU'LL HEAR**	**TOOLS YOU'LL HEAR**	**TOOLS YOU'LL HEAR**	**TOOLS YOU'LL HEAR**	**TOOLS YOU'LL HEAR**
Notion AI · ChatGPT ReadAI	Cursor · Claude Code GPT Codex · Copilot · Devin	GitHub Copilot Review CodeRabbit	Meticulous · Momentic Playwright + Claude	Sentry AI · Cursor 'fix' Runbook agents	Dependabot · Renovate Claude Code w/ shell
⚠ **WATCH OUT**	⚠ **WATCH OUT**	⚠ **WATCH OUT**	⚠ **WATCH OUT**	⚠ **WATCH OUT**	⚠ **WATCH OUT**
invented constraints	shipped ≠ value	amplifies weak culture	both sides of assertion	confidently wrong	kubectl + bad idea

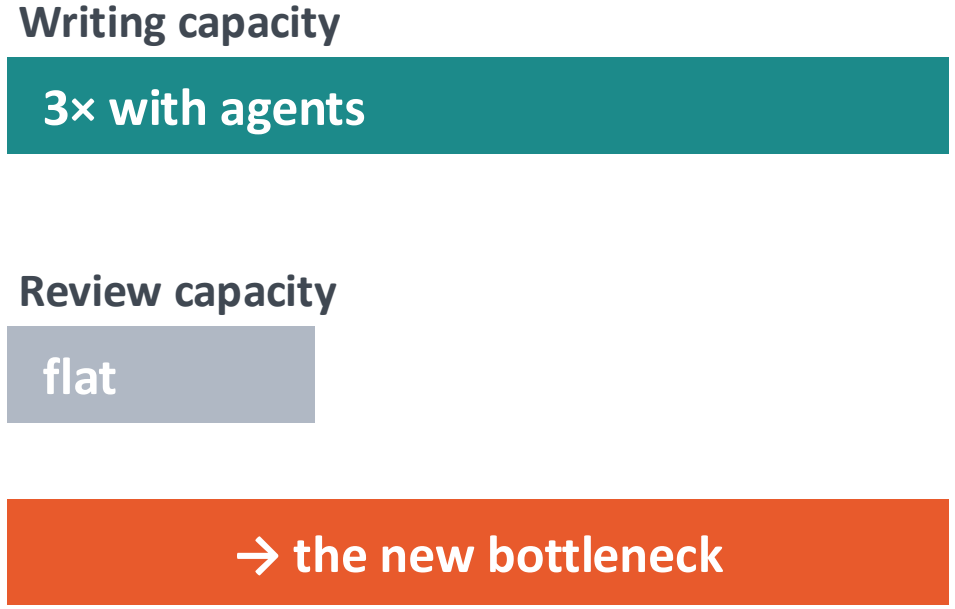
Zoom in: Code Review

The stage where the ROI conversation actually lives.

The argument

1. When agents write more code, review becomes the bottleneck.
2. Review is where quality, security, and standards enter your codebase.
3. Hence, a high-priority agent is the reviewer, not the coder.

Capacity shift



The PR-review agent uses same seven primitives, different job.

The PR-review agent: same seven pieces

Compare pill-for-pill with slide 22. Different application. Same primitives.

1 LLM	2 Tools	3 RAG	4 Skills	5 Memory	6 LangGraph	7 MCP + A2A
the reasoner reading the diff	git diff · run tests static analysis · coverage	your codebase past PRs · ADRs	review checklist as versioned playbook	team preferences (who hates what)	sub-agents: security · style · tests · performance	MCP → GitHub / CI A2A → ComplianceAgent

diff → plan → call tools → observe → post review comments → hand off to ComplianceAgent

One matrix: same primitives, every stage

Tool / concept	Reqs	Code	Review	Test	Debug	DevOps
Cursor / Claude Code	●	●	●	●	●	–
MCP servers	●	●	●	●	●	●
RAG (internal docs)	●	–	●	–	●	–
Skills	●	●	●	●	●	–
LangGraph flows	●	–	●	●	●	–
Sub-agents / A2A	●	●	●	●	●	–

The 'Cursor vs Copilot vs Claude Code' argument matters less than which primitives your team is fluent in.

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Part 4: Where humans stay indispensable

Three things agents remain bad at

01

Deciding what to build

An agent will happily build the wrong thing beautifully. Prioritisation is a judgment call about people and money.

02

Owning a decision

When it's wrong, nobody's fired. Accountability doesn't survive being handed to a model.

03

Novel taste

Agents interpolate from training data. Anything genuinely new — a brand voice, a bold architecture — is on you.

The engineers who thrive shift from typing code to reviewing, testing, and directing agents.

Human-in-the-loop: pick a pattern

HIGH RISK

Approve-to-act

Agent proposes. Human clicks yes.

USE WHEN

Anything that costs money, sends email, or writes to prod.

BULK WORK

Sample-and-audit

Agent acts freely. Human reviews N% of outputs.

USE WHEN

High-volume, low-stakes work.

MIXED

Escalate-on-uncertainty

Agent handles the easy 80%. Flags the hard 20%.

USE WHEN

Support, triage, classification.

The guardrail is you. Pick the pattern per workflow; don't default.

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The leverage isn't writing code faster.

It is redesigning the work to move humans off repetitive tasks and onto the judgment, taste, and accountability that agents still can't fake.

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Thank you!